

Python ALA-2= What is List and Dictionary:

1. What is a List in Python?

Definition:

A **List** is an ordered, mutable collection of elements that allows duplicate values and is written using square brackets `[]`.

Syntax: `list_name = [value1, value2,`

`value3]` **Example:**

```
numbers = [10, 20, 30, 40]
```

```
print(numbers) Output:
```

```
[10, 20, 30, 40]
```

Answer:

List stores multiple values in a single variable and allows modification.

2. What is a Dictionary in Python?

Definition:

A **Dictionary** is an unordered collection of key-value pairs that is mutable and written using curly braces `{}`.

Syntax: `dict_name = {key1:value1,`

`key2:value2}` **Example:**

```
student = {"name": "Rahul", "age": 18}
```

```
print(student) Output:
```

```
{'name': 'Rahul', 'age': 18}
```

Answer:

Dictionary stores data in key-value format.

3. Difference between List and Dictionary?

Definition:

List stores values in order, Dictionary stores values in key-value pairs.

Example:

```
# List
fruits = ["apple", "banana", "mango"]

# Dictionary fruits_dict = {"a": "apple",
"b": "banana"} Answer:
```

List	Dictionary
Ordered	Key-Value pair
Uses []	Uses {}
Index based	Key based
Allows duplicate values	Keys must be unique

4. How do you access elements in a List?

Definition:

List elements are accessed using index number.

Example:

```
numbers = [10, 20, 30, 40]

print(numbers[0])
print(numbers[2]) Output:

10
30
```

Answer:

Index starts from 0.

5. How do you access values in a Dictionary?

Definition:

Dictionary values are accessed using keys.

Example:

```
student = {"name": "Rahul", "age": 18}
print(student["name"])
print(student["age"])
```

Output:

Rahul 18

Answer:

Keys are used instead of index.

6. How to add elements in List and Dictionary?

Definition:

Elements can be added using append() in list and new key in dictionary.

Example:

```
# List numbers = [10, 20]
numbers.append(30) #
Dictionary student =
{"name": "Rahul"}
student["age"] = 18
print(numbers)
print(student)
```

Output:

```
[10, 20, 30]
{'name': 'Rahul', 'age': 18}
```

7. How to remove elements from List and Dictionary?

Definition:

Elements can be removed using remove() or pop().

Example:

```
# List
numbers = [10, 20, 30]
numbers.remove(20)

# Dictionary
```

```
student = {"name": "Rahul", "age": 18}
student.pop("age")
print(numbers)
print(student)
```

Output:

```
[10, 30]
{'name': 'Rahul'}
```

8. Can List and Dictionary store different data types?

Definition:

Yes, both can store multiple data types.

Example:

```
# List
data = [10, "Rahul", 5.5]

# Dictionary student
= {
    "name": "Rahul",
    "age": 18,
    "marks": 90
} print(data)
print(student)
```

Output:

```
[10, 'Rahul', 5.5]
{'name': 'Rahul', 'age': 18, 'marks': 90}
```

9. How to loop through List and Dictionary?

Definition:

Loop is used to iterate elements.

Example:

```
# List
numbers = [10, 20, 30]
for n in
numbers:
    print(n)

# Dictionary
student = {"name": "Rahul", "age": 18}
```

```
for key, value in
student.items():    print(key,
value) Output:
```

```
10
20 30 name
Rahul
age 18
```

10. When to use List and Dictionary?

Definition:

Use List when order and indexing is needed.

Use Dictionary when key-value mapping is needed.

Example:

```
# List
marks = [90, 80, 70]

# Dictionary
student_marks = {
"math": 90,
    "science": 80,
    "english": 70 }
```

Answer:

- List → Store multiple items in order
- Dictionary → Store data with keys
- List → Index based
- Dictionary → Key based